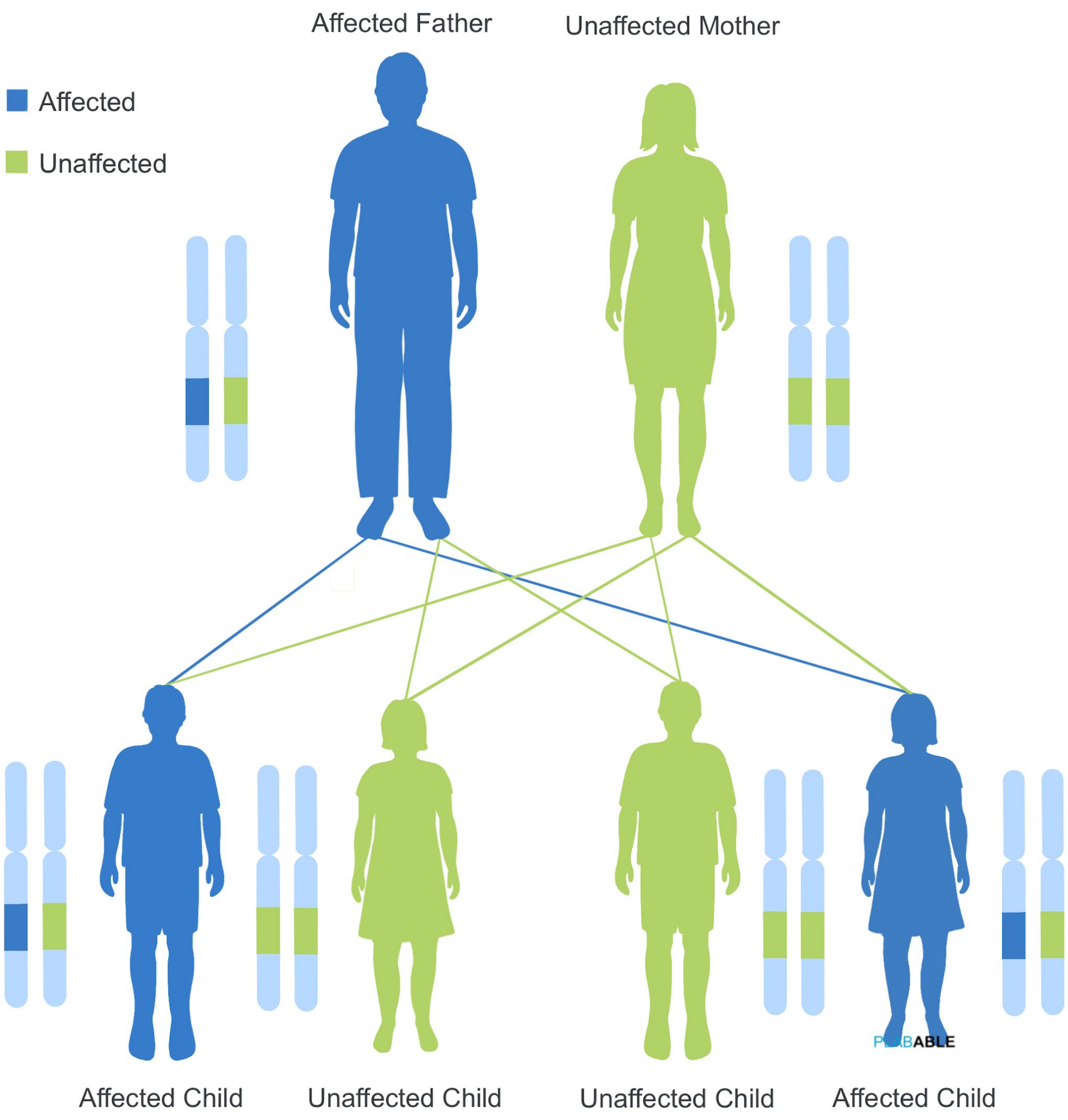
PLABABLE

GEMS

VERSION 1.2

GENETICS

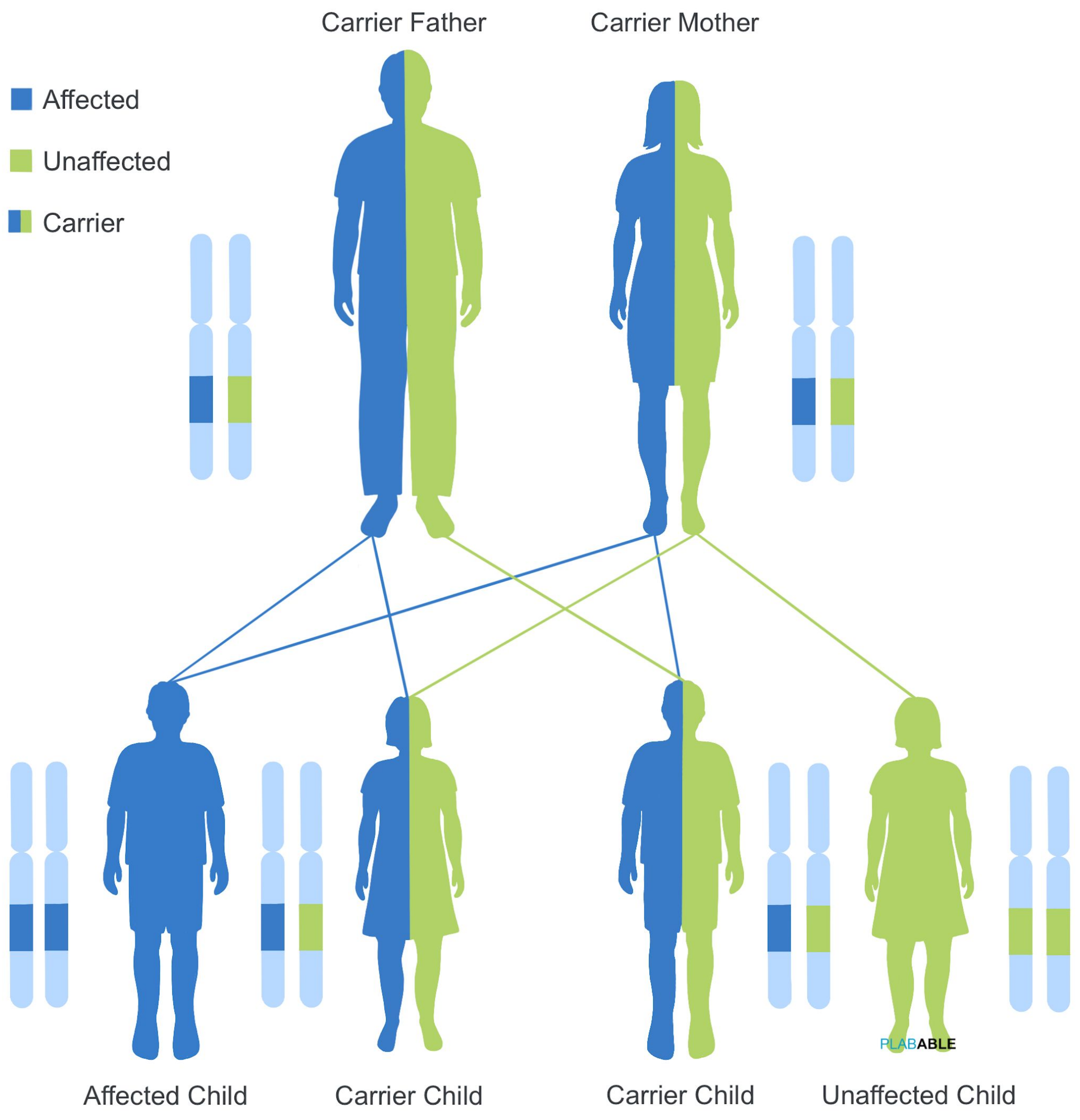
Autosomal Dominant Inheritance Affected Father



Autosomal dominant conditions

- Polycystic kidney disease
- Huntington’s
- Neurofibromatosis

Autosomal Recessive Inheritance

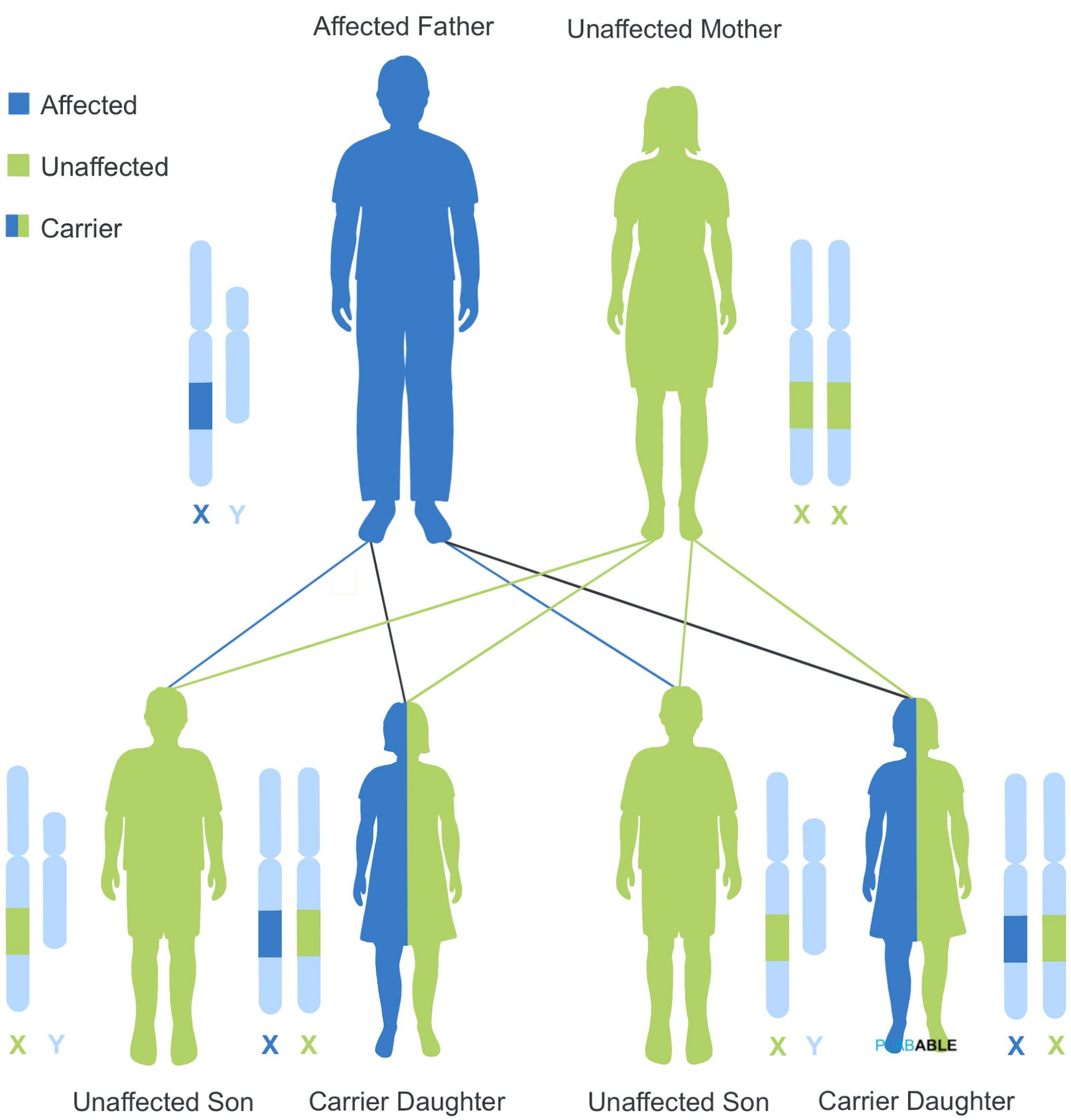


Autosomal recessive conditions

- Cystic fibrosis
- Thalassemia
- Sickle cell anaemia
- Wilson's disease
- Congenital adrenal hyperplasia

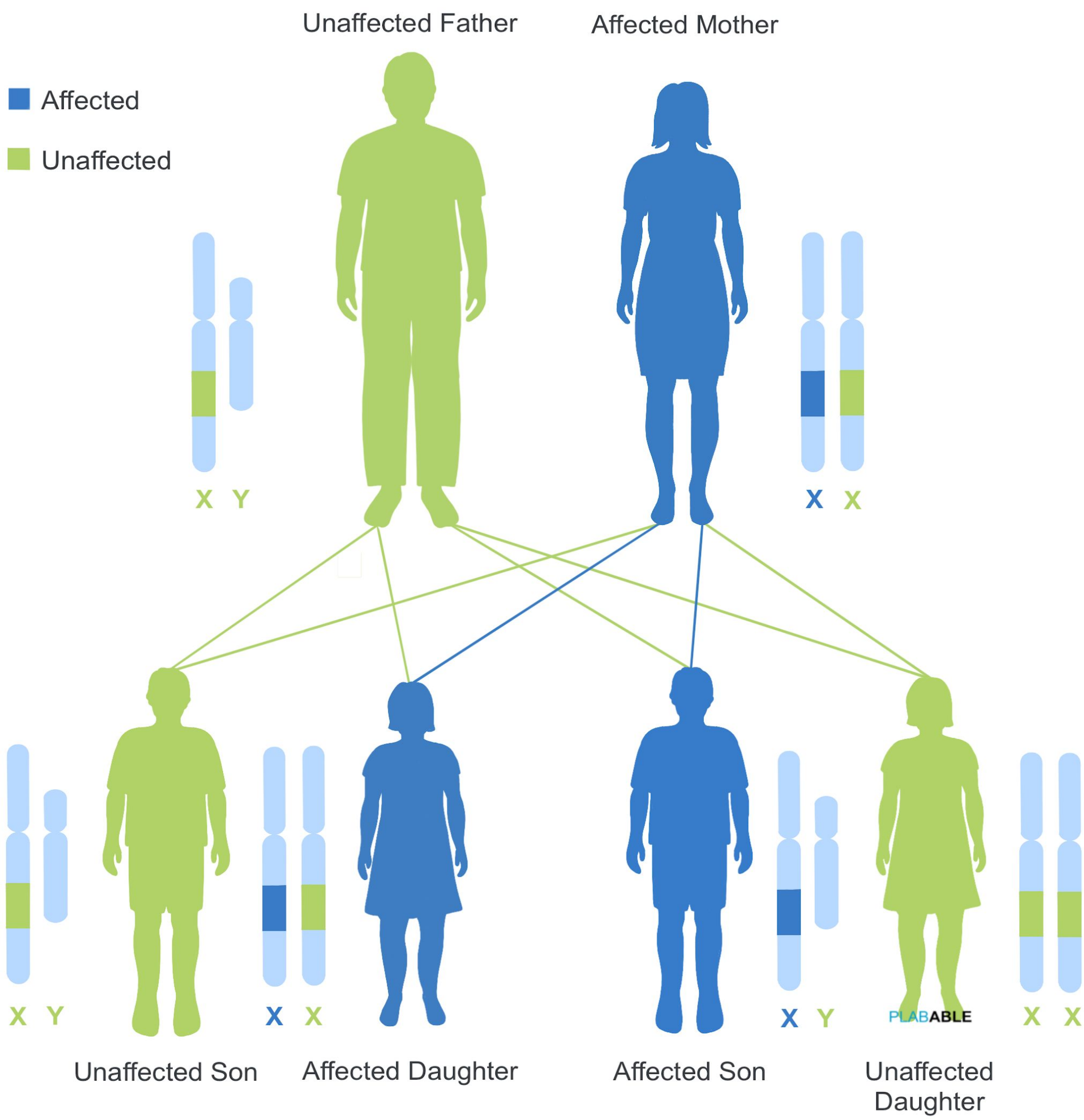
PLABABLE

X-Linked Recessive Inheritance (Affected Father)



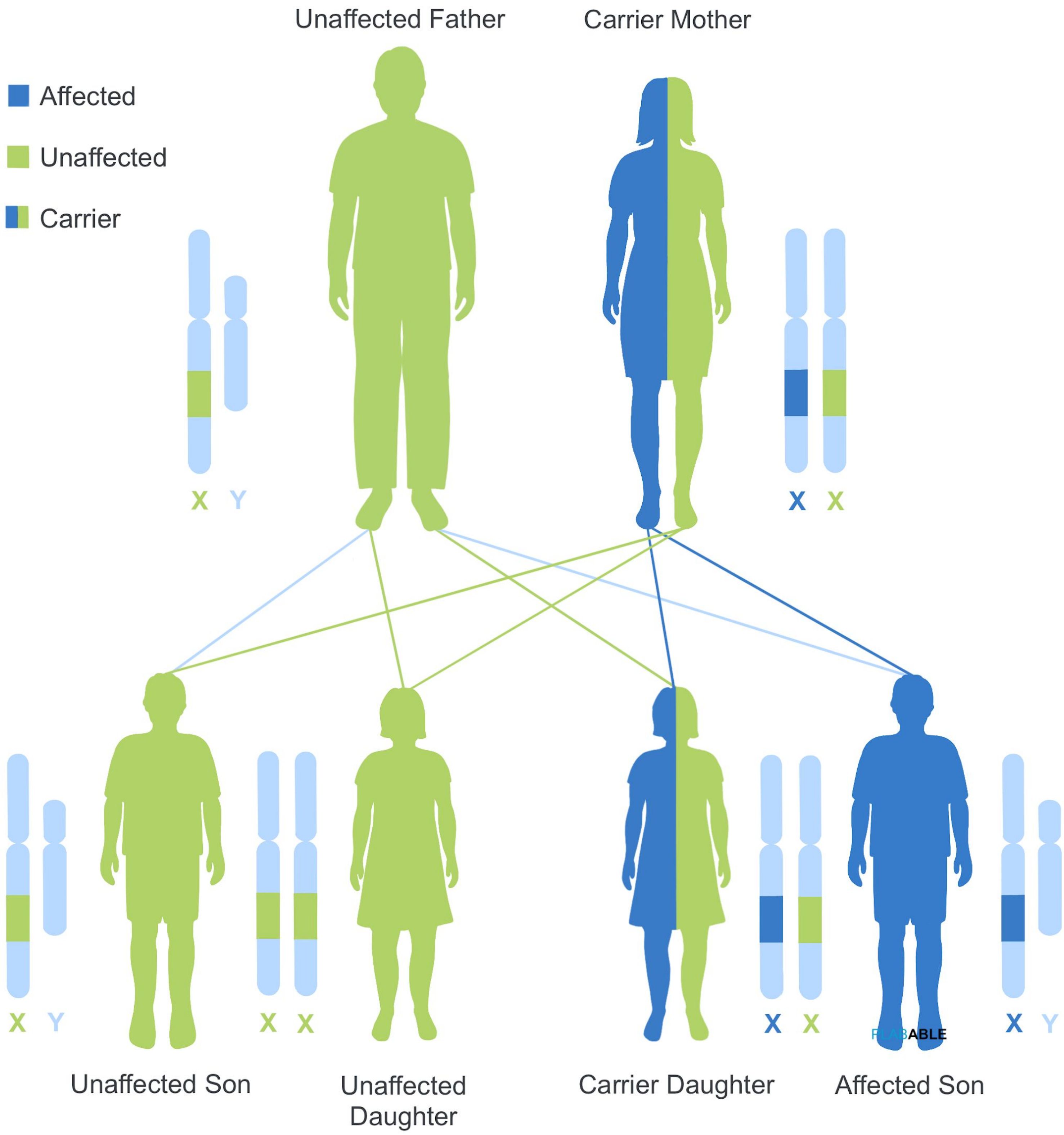
X-Linked Recessive Inheritance

X-Linked Dominant Inheritance (Affected Mother)



X-Linked Dominant Inheritance

X-Linked Recessive (Carrier Mother)



X-Link Recessive - Carrier Mother

Modes of Genetic Inheritance

Autosomal recessive:

If both *parents are carrier:*

- **25%** chance of a child to be **affected**
- **50%** chance of a child to be **a carrier**

e.g. cystic fibrosis, congenital adrenal hyperplasia, sickle cell anaemia and thalassemia

Autosomal dominant:

If *one of the parents is affected*:

- **50%** chance of a child to be **affected**
- **25%** chance of a **grandchild** to be **affected**

e.g. Huntington's diseases, neurofibromatosis, autosomal dominant polycystic kidney disease

X-Linked recessive:

If *one of the parents is carrier*:

- **50%** chance of a **male child** to be **affected**

e.g. Duchenne muscular dystrophy (DMD), haemophilia

Careful: Is the question is asking **carrier** or **affected**?

Cystic Fibrosis

It is caused by autosomal recessive mutation in **CFTR gene** (cystic fibrosis transmembrane conductance regulator gene)

↑ Viscosity and thickness of body's secretion

+

High chloride in the skin

It is an **autosomal recessive** disease

→ If both parents are carriers then the children:

- 25% = **Unaffected** (completely healthy and does not carry mutated gene)
- 50% = **Carrier** (carry mutated gene but not have the disease)
- 25% = **Affected** (have the disease)

Symptoms of Cystic Fibrosis

- **Salty skin** - parents notice when kissing child
- **Finger clubbing**
- **Frequent cough, wheeze, SOB, pneumonia and sinusitis -** thick secretion & mucus accumulates in the 'alveoli': Recurrent repetitive cough, respiratory failure, bronchiectasis as long term complications
- Congenital absence of vas deferens → **Infertility in males**
- **Meconium ileus** (first stool that is passed by a newborn will not pass due to thickness)
- **Big appetite but poor weight gain, failure to thrive, steatorrhoea** →Thick secretion blocking pancreatic duct resulting in no pancreatic enzymes released,↓ fat and protein absorption

Cystic Fibrosis Management

Airway clearance techniques

Immunomodulatory agent:

- For patients with deteriorating lung function
- Or repeated pulmonary exacerbation

→ Long term azithromycin

→ Or oral corticosteroid if no improvement with azithromycin, stop azithromycin

Mucolytic agents:

- rhDNase ± hypertonic sodium chloride or hypertonic sodium chloride alone
- Mannitol dry powder
- Lumacaftor–ivacaftor

Treating pulmonary infections:

- Antibiotics or antifungals depends on types of infection and sensitivity

Cystic fibrosis

Brain trainer:

A 3 year old boy presents with rectal prolapse, failure to thrive, bulky stools and a repetitive cough over the last few months. What is the most appropriate initial test ?

➔ **Sweat test**

Polycystic Kidney Disease

It is an **autosomal dominant** disease

→ 50% affected in children if one parent has it

Features:

- **Abdominal, flank, back pain**
- **Haematuria**
- **Hypertension** (early manifestation)
  → ACE-inhibitors or ARB to control BP
- **Cerebral aneurysms**

Polycystic kidney disease is associated with cerebral aneurysms

→ Once ruptured can cause **subarachnoid haemorrhage**

Turner Syndrome

Facts:

- It affects **female** only
- Most have normal intelligence but **some have learning difficulties**
- Advanced age mother is **NOT** a risk factor for Turner's syndrome
- Affected patients are **infertile**, however can conceive by assisted reproductive techniques

Features:	
Short stature	Ovarian failure (1yr amenorrhea)
Short webbed neck	Impaired pubertal growth
Widely spaced nipples	Bicuspid aortic valve

Management:

- **Human growth hormone** (GH) is used during childhood to **increase height**
- **Oestrogen replacement** is used during childhood:

→ Enhance **breast and hip development**

→ **Prevent osteoporosis**

Turner syndrome

Brain trainer:

What technique is available for women with Turner’s syndrome to conceive ?

➔ **Assisted reproduction techniques**

Prader Willis Syndrome

Facts:

- It is a genetic disease

→ **Deletion** of some genes of the **paternal chromosome** #15

During neonatal period:

- **Hypotonia (floppy baby)**
- Difficult to feed due to thin upper lip and downturned mouth
- Short extremities
- Almond-shaped eyes

During childhood period:

- **Excessive eating (hyperphagia)**
- Obese and short
- Learning difficulties
- Growth abnormalities
- Self-injurious behaviour

Angelman Syndrome

Facts:

- It is a genetic disease

→ **Lack of expression** of maternally inherited UBE3A gene in the brain (CHR15)

During neonatal period:

- Normal head circumference at birth
- Developmental delay is apparent by 6 months

During childhood period:

- **Functionally severe developmental delay**
- **Delay disproportionate head circumference growth**
- Jitteriness from 6 months with irregulars, coarse movement that prevent walking and reach for objects
- Speech impairment with no or minimal use of words

Duchenne Muscular Dystrophy

Facts:

- It is an X-linked recessive diseases
- X-linked recessive only affects male
  → If one of the parents is carrier, then a male child will have 50% to inherit the gene

Features:

- 4-8 yo boy **started to walk late** i.e. **>18 months** rather than 12 months
- **Gower's sign** → use his hands to push on his legs to stand
- **Proximal muscle weakness**
- **Waddling gait** → Cannot run
- ↑ **Creatine kinase**, ALT, AST
- $\pm$ Respiratory and/or cardiac manifestation

Diagnosis:

- Initial test → creatine kinase
- Muscle biopsy
- Genetic testing (obligatory after +ve muscle biopsy)

DMD has a mutation defect in *dystrophin protein* which lies in *striated muscles*

Other X-linked recessive: **Haemophilia**

Fragile X syndrome

Brain trainer:

What is the mode of inheritance for Fragile X-syndrome ?

- ➔ **X-linked dominant**

Becker Muscular Dystrophy

Brain trainer:

What is the mode of inheritance for Becker Muscular Dystrophy ?

➔ **X-linked recessive**

Trisomy Syndrome

Trisomy 13 Patau's syndrome	Trisomy 18 Edward's syndrome	Trisomy 21 Down syndrome
Prominent calcaneus	Prominent calcaneus	Space between 1st and 2nd toe
Cleft lips and palate	Prominent occiput	Abnormally shaped ear
Microcephaly (small head)	Microcephaly (Small head)	Small mouth with enlarged tongue
Microphthalmia (small eyes)	Micrognathia (Small jaw)	Upward slant to the eyes
Polydactyly (multiple fingers)	Hands clenched into fists with overriding fingers	Small nose with depressed nasal bridge and joint laxity
		• Single deep crease across palm • Duodenal atresia • Double bubble sign on chest X-ray

DiGeorge Syndrome

Brain trainer:

A 3 month old baby with recurrent infections, feeding difficulties, dysmorphic face and cleft palate. X-ray shows absent thymic shadow. What is the most likely diagnosis?

➔ **DiGeorge syndrome**

Remember CATCH-22

- Cardiac abnormality
- Abnormal facies
- Thymic aplasia
- Cleft palate
- Hypocalcemia/Hypoparathyroidism
- Chromosome 22 abnormality

Congenital Adrenal Hyperplasia

Facts:

- It is an **autosomal recessive** disease → If both parents are carrier, then **25%** chance in child will be affected
- **21 hydroxylase deficiency** is the most common form

Features:

- ± Cortisol deficiency
- ± Aldosterone deficiency
- ± Androgen excess

Female:

- Ambiguous genitalia

Male:

- Penile enlargement
- Hyperpigmentation

Infant male:

- Salt wasting due to aldosterone deficiency
- Vomiting
- Weight loss
- Lethargy
- Dehydration
- Hyponatraemia
- Hyperkalaemia
- 11-β-hydroxylase deficiency

PLABABLE

Klinefelter Syndrome

Facts:

- It is also known as 47, XXY males
- It affects **boys and males** only

Features:

G-FELTER

- **G**ynecomastia
- **F**acial Hair - low
- **E**strogen is high and testosterone is low
- **L**ong limbs
- **T**all and slim
- **E**levated FSH and LH
- **R**age - aggressive behaviour

+

Hypogonadism:

- Small Testes
- Azoospermia (no sperms in seman)
- Male infertility

Diagnosis:

Karyotyping = chromosome analysis (47 XXY)

Huntington’s Disease

Facts:

- It is an autosomal dominant disease
  → If one parent is affected then 50% chance of child will be affected

Features:

- Changes of personalities, self-neglect, clumsiness - early sign
- Progressive cognitive impairment
  e.g. memory loss, poor concentration
- Chorea - involuntary writhing jerky movement of limbs
- Dystonia
- Rigidity
- Dementia

Remember

Jerky involuntary movement

Cognitive impairment + Family history

Anticipation = symptoms in next generation appears earlier and earlier (at young age)

Investigating Potential Genetic Diseases

Before pregnancy:

- **Preimplantation genetic diagnosis (PGD)**

1. Fertilization is done in vitro (laboratory)
2. Embryos are tested for genetic abnormalities
3. 1-2 unaffected embryos are then implanted to uterus

→ Suitable to check for possible **serious genetic diseases** such as **autosomal recessive**

→ To check for **disease with strong family history** and **if parent is carrier**

Week 11-14 of pregnancy:

- **Chorionic villus sampling (CVS)**

→ a small sample of placenta is tested

Week 15-20 of pregnancy:

- **Amniocentesis**

→ A small sample of amniotic fluid is tested

Genetic screening

Brain trainer:

A woman with a family history of cystic fibrosis is planning to have a baby and wants to receive the earliest possible test for this disease. What test would you recommend?

➔ **Preimplantation genetic diagnosis**

Genetic screening

Brain trainer:

A 41 year old pregnant woman is 17 weeks pregnant and is worried about a chromosomal abnormality. What is the most definitive investigation at this stage?

➔ **Amniocentesis**

Ehlers-Danlos Syndrome

Ehlers-Danlos syndrome is a collagen disorder

Features:

Hyperplasticity of skin

Hypermobility of joints

±
Blue sclera

Neurofibromatosis (NF)

Facts:

- It is an autosomal dominant disease

→ If 1 parent affected = 50% chance of child affected

Diagnostic criteria or features:

- >6 cafe au lait macules (>0.5cm in children or >1.5cm in adults)
- >2 cutaneous/subcutaneous neurofibromatosis or one plexiform neurofibromatosis
- Axillary or groin freakling
- Optic pathway glioma
- >2 Lisch nodules (iris hamartomas seen on slit lamp examination)
- Bone dysplasia (sphenoid wing dysplasia, bowing of long bone ± pseudarthrosis)
- 1st degree relative with NF Type 1

Neurofibromatosis (NF)

Type 1 NF

Present more with **skin lesions**

- Cafe-au-lait spots
- Axillary/groin freckles
- Iris hamartomas
- Scoliosis

Type 2 NF

Present more with **CNS tumour**

- BIlateral acoustic neuroma
- Multiple intracranial schwannomas
- Meningiomas

Ovarian Cancer

Facts:

- Woman with strong 1st or 2nd degree family history has higher risk
- Presence of **BRCA1 and 2** genes increases susceptibility

1. Do **pelvic ultrasound**

2. **Genetic counselling** if unremarkable for full assessment of risk

3. Offer **prophylactic salpingo-oophorectomy** if high risk

If there are manifestation:

Perform ultrasound
+
Check for Ca-125 Level

PLABABLE

BRCA1

Brain trainer:

What is the mode of inheritance for BRCA1 gene mutation?

➔ **Autosomal dominant with incomplete penetrance**

Alport Syndrome

Facts:

- It is a **X-linked disease**
- Father passes on Y-chromosome, mother passes on X-chromosome
- If **father is affected** but **mother is not,** chance of male child affected is nearly 0%

Features:

- Kidney Disease:
  → Haematuria + proteinuria
  → End stage renal disease
- Hearing Loss
  → Sensorineural hearing loss (SHNL)
- Eyes abnormalities

Haemolytic disease of the newborn

Brain trainer:

A pregnant woman who is blood type O negative suffered with haemolytic disease of a newborn when she was born. What is the likelihood of her first born having haemolytic disease of a newborn?

➔ **0%**

Remember haemolytic disease of a newborn is NOT a genetic condition!

Genes & Associated Diseases

Gene	Disease
BCL2	B cell lymphoma
Apolipoprotein E (APOE)	Alzheimer's
FBN1	Marfan Syndrome
NOTCH1	T-cell leukaemia
HOXB13	Prostate cancer

Image Attribution

https://commons.wikimedia.org/wiki/File:Genetic_engineering_logo.png